MoCafe v2.00 — Physics and Algorithms

Abstract

This note documents the physics and the numerical algorithms behind MoCafe, the parallel Monte-Carlo dust radiative-transfer code. It develops the radiative transfer problem and the Monte-Carlo solution (photon packets, free-path sampling, forced first scattering, weighting, and Russian roulette), the peeling-off (next-event) estimator, the scattering physics (Henyey–Greenstein and the polarized Mueller-matrix path), panchromatic transport by grey rescaling, the Lucy (1999) path-length estimator for the radiation field, the two dust-emission solvers (the Lucy iteration coupled to the SEDUST grain library with quantum-mechanical stochastic heating, and the Bjorkman & Wood immediate-reemission scheme), the Modified Random Walk, the galaxy source models, the random-sampling routines (inversion, rejection, and Walker's alias method), the quasi-random (Owen-scrambled Sobol) launch option, and the HEALPix interior observer. Derivations are given at the level needed to read and modify the code; the companion user guide covers installation, the input reference, and the outputs.

Contents

1. The radiative transfer problem

1.1 The transfer equation

The specific intensity $I_\lambda(\mathbf{r},\hat{\mathbf{n}})$ of radiation of wavelength λ at position $\mathbf{r}$ travelling in direction $\hat{\mathbf{n}}$ obeys the radiative transfer equation

$$\hat{\mathbf{n}}\cdot\nabla I_\lambda = -\kappa_{\rm ext}(\mathbf{r},\lambda)\,I_\lambda + \kappa_{\rm ext}(\mathbf{r},\lambda)\,S_\lambda(\mathbf{r},\hat{\mathbf{n}}), \tag{1}$$

where $\kappa_{\rm ext} = \kappa_{\rm sca} + \kappa_{\rm abs}$ is the extinction coefficient (per unit length), $\kappa_{\rm sca}$ and $\kappa_{\rm abs}$ the scattering and absorption coefficients, and S_λ the source function. For a dusty medium the source function has a scattering part and a thermal-emission part,

$$S_\lambda(\mathbf{r},\hat{\mathbf{n}}) = \frac{\kappa_{\rm sca}}{\kappa_{\rm ext}}\int \frac{{\rm d}\Omega'}{4\pi}\,\Phi(\hat{\mathbf{n}},\hat{\mathbf{n}}')\,I_\lambda(\mathbf{r},\hat{\mathbf{n}}') + \frac{\kappa_{\rm abs}}{\kappa_{\rm ext}}\,B_\lambda\big(T_{\rm d}(\mathbf{r})\big), \tag{2}$$

with Φ the (normalized) scattering phase function, $a \equiv \kappa_{\rm sca}/\kappa_{\rm ext}$ the single-scattering albedo, and $B_\lambda(T_{\rm d})$ the thermal emission of dust at temperature $T_{\rm d}$. Defining the optical depth along a ray by ${\rm d}\tau_\lambda = \kappa_{\rm ext}\,{\rm d}\ell$, the formal solution of Eq. (1) is

$$I_\lambda(\tau_\lambda) = I_\lambda(0)\,e^{-\tau_\lambda} + \int_0^{\tau_\lambda} S_\lambda(\tau')\,e^{-(\tau_\lambda-\tau')}\,{\rm d}\tau', \tag{3}$$

i.e. attenuated incident light plus the attenuated contribution of every source point along the ray.

1.2 Why Monte Carlo

Equations (1)–(2) couple every direction and position (scattering redistributes light in angle; thermal emission couples wavelengths through the temperature). Rather than discretizing the integro-differential operator, MoCafe evaluates the physical content of Eq. (3) *statistically*: it follows a large number of photon packets whose random histories — emission, propagation over an exponentially distributed optical depth, scattering into a new direction, absorption — reproduce the transfer equation in expectation. Integrals over the phase function, the size distribution, and the wavelength grid become sums over sampled events, so the cost scales with the number of packets rather than with the dimensionality of the problem, and complex three-dimensional geometries (Cartesian, clumpy, AMR) are handled by the same estimator.

2. Monte-Carlo photon transport

2.1 Photon packets and weights

MoCafe transports photon *packets*: weighted bundles of monochromatic photons. A packet carries a position $\mathbf{r}$, a propagation direction $\hat{\mathbf{k}}$, a dimensionless weight w, a wavelength bin λ, a Stokes vector (I,Q,U,V) when polarization is on, and—in SED/emission mode—a physical luminosity $L_{\rm p}$ [erg s^{-1}] so that energy budgets are absolute. The weight absorbs all the variance-reduction bookkeeping (forced scattering, albedo, scan reweighting); the packet contributes $wL_{\rm p}$ wherever it is tallied.

2.2 Sampling the free path

The probability that a packet reaches optical depth τ without interacting is $e^{-\tau}$, so the interaction optical depth has probability density $p(\tau) = e^{-\tau}$ and cumulative $P(\tau) = 1 - e^{-\tau}$. Inverting $P(\tau) = \xi$ for a uniform deviate $\xi \in [0, 1)$ gives the standard free-path sample

$$\tau = -\ln(1 - \xi). \tag{4}$$

The sampled τ is converted to a physical displacement by integrating $\kappa_{\rm ext}$ along the ray until the accumulated optical depth equals τ. The three media do this with a shared interface: Amanatides–Woo voxel traversal on the Cartesian grid, a ray–sphere intersection followed by a DDA walk over the clump acceleration grid in the clumpy medium, and an octree neighbour walk on the AMR grid. Because scattering and peeling-off call these tracers through a procedure pointer, they are written once and are grid-agnostic.

2.3 Forced first scattering and reduced weight

In an optically thin medium most packets would escape without interacting, which wastes samples. Setting `par%use_reduced_wgt` forces the first interaction to occur inside the grid. Let τ_0 be the total optical depth from the birth point to the far edge along $\hat{\mathbf{k}}$. Restricting the free-path distribution to $[0, \tau_0]$ and renormalizing gives the truncated density $p(\tau) = e^{-\tau}/(1 - e^{-\tau_0})$, whose inverse is

$$\tau = -\ln\left[1 - \xi\,(1 - e^{-\tau_0})\right], \qquad \tau \in [0, \tau_0]. \tag{5}$$

To keep the estimator unbiased the packet weight is multiplied by the probability that it would have interacted at all,

$$w \leftarrow w(1 - e^{-\tau_0}), \tag{6}$$

and the complementary fraction $e^{-\tau_0}$ — the probability of escaping without scattering — is carried by the direct (unscattered) peel-off toward each observer, so no energy is lost. This is why the $n_{\rm scatt} = 0$ step is special-cased in the transport loop.

2.4 Absorption, weighting, and Russian roulette

At an interaction MoCafe uses *albedo weighting* rather than a stochastic absorb/scatter decision: the packet always scatters, and the absorbed fraction is accounted continuously (the scattered peel carries a factor a; the absorbed energy is deposited into the radiation-field tally, §6.). This removes the variance of the absorb/scatter coin. A packet whose weight has decayed below a cutoff is subjected to Russian roulette: with survival probability q it continues with weight w/q, otherwise it is terminated. Roulette keeps the estimator unbiased while preventing unbounded tracking of negligible packets, and is used both in the transport loop and inside the Modified Random Walk (§9.).

3. The peeling-off (next-event) estimator

3.1 Derivation

Waiting for packets to reach a distant observer by chance is hopelessly inefficient. Instead, at every emission and scattering site MoCafe adds the *expected* contribution the packet would make toward each observer — the peeling-off or next-event estimator. Consider a scattering at position $\mathbf{r}$ with incoming direction $\hat{\mathbf{k}}$. The joint probability that the packet scatters into the small solid angle $\mathrm{d}\Omega_o$ about the observer direction $\hat{\mathbf{n}}_o$ *and* then escapes to the observer without further interaction is

$$a\,\frac{\Phi(\hat{\mathbf{k}}\cdot\hat{\mathbf{n}}_o)}{4\pi}\,\mathrm{d}\Omega_o \times e^{-\tau_o}, \tag{7}$$

where a is the albedo, Φ the normalized phase function, and τ_o the optical depth from $\mathbf{r}$ to the observer along $\hat{\mathbf{n}}_o$. For an observer of collecting area $\mathrm{d}A$ at distance d_o, $\mathrm{d}\Omega_o = \mathrm{d}A/d_o^2$, so the image-plane contribution per unit area is

$$\Delta I_o = \frac{w\,a\,\Phi(\hat{\mathbf{k}}\cdot\hat{\mathbf{n}}_o)}{4\pi}\,\frac{e^{-\tau_o}}{d_o^2}. \tag{8}$$

The $1/d_o^2$ is the geometric dilution, $e^{-\tau_o}$ the attenuation to the observer (computed by ray-tracing to the grid edge), and the phase-function factor the probability of redirection. Summing Eq. (8) over all scattering events of all packets builds the scattered image with far lower variance than collecting escaping packets, because *every* event contributes to *every* observer.

3.2 Direct and scattered peels

The unscattered (direct) light is peeled separately: a source packet contributes its birth luminosity attenuated by the optical depth from the source to the observer, forming the `Direct` image (and `Direct0`, the unattenuated version, when requested). Subsequent scatterings feed the `Scattered` image through Eq. (8). For an internal all-sky observer (§12.) the same expected contribution is binned into the HEALPix pixel of the arrival direction instead of a tangent-plane pixel. All peels go through the `raytrace_to_edge` procedure pointer, so the Cartesian, clumpy, and AMR media share identical scattering and peeling code; the default binding keeps the Cartesian output bit-identical across refactors.

4. Scattering

4.1 Henyey–Greenstein

The default phase function is Henyey–Greenstein with asymmetry parameter $g = \langle\cos\theta\rangle$,

$$\Phi_{\rm HG}(\mu) = \frac{1-g^2}{(1+g^2-2g\mu)^{3/2}}, \qquad \mu = \cos\theta, \tag{9}$$

normalized so that $\frac{1}{2}\int_{-1}^{1}\Phi_{\rm HG}\,\mathrm{d}\mu = 1$. Its cumulative distribution inverts in closed form, so the scattering cosine is sampled directly from a uniform deviate ξ:

$$\mu = \frac{1}{2g}\left[1+g^2-\left(\frac{1-g^2}{1-g+2g\xi}\right)^2\right] \quad (g\neq 0), \tag{10}$$

with the isotropic limit $\mu = 2\xi-1$ for $g=0$. The azimuth is uniform, and the new direction is built by rotating $\hat{\mathbf{k}}$ by (θ,φ).

4.2 Polarization: Stokes vectors and the Mueller matrix

When a scattering-matrix table is supplied (`par%scatt_mat_file`) the code carries the Stokes vector $\mathbf{S} = (I, Q, U, V)^\top$ and replaces the scalar phase function by the 4×4 Mueller matrix $\mathbf{M}(\theta)$ of the grains. A scattering event proceeds in three steps: (i) rotate $\mathbf{S}$ into the scattering plane by the Stokes rotation $\mathbf{R}(\psi_1)$; (ii) apply the Mueller matrix, $\mathbf{S}' = \mathbf{M}(\theta)\,\mathbf{R}(\psi_1)\,\mathbf{S}$; (iii) rotate the result into the new meridian frame by $\mathbf{R}(\psi_2)$. The scattering angle is drawn from the tabulated $S_{11}(\theta)$ (the intensity phase function) with an alias table (§11.) for $\mathcal{O}(1)$ sampling; because S_{11} already encodes the angular dependence, the azimuth is drawn from the S_{12}-modulated distribution to respect the incident polarization. The peel-off of Eq. (8) then uses the matrix-weighted phase function evaluated for the observer direction, so the polarized images (Q, U, V planes) are next-event estimates just like I. With Henyey–Greenstein, `par%use_stokes` is forced off.

5. Panchromatic (SED) transport

5.1 Grey rescaling

In SED mode a packet is assigned a wavelength bin at birth and keeps it until absorption. Storing a separate opacity grid per wavelength would multiply the memory by the number of bins; instead MoCafe stores the density structure — and hence the optical depth — at a single reference wavelength $\lambda_{\rm ref}$ and rescales each packet by the wavelength-dependent opacity ratio (Jonsson 2006),

$$s_{\rm ext}(\lambda) = \frac{\kappa_{\rm ext}(\lambda)}{\kappa_{\rm ext}(\lambda_{\rm ref})}, \qquad \tau(\lambda) = s_{\rm ext}(\lambda)\,\tau_{\rm ref}. \tag{11}$$

The ray tracers integrate $\tau_{\rm ref}$ and multiply by the packet's $s_{\rm ext}$; the free-path draw of Eq. (4) is done in $\tau_{\rm ref} = \tau(\lambda)/s_{\rm ext}$. Because $s_{\rm ext}$ is a single scalar per packet, all three tracers are shared with the monochromatic path and the memory does not grow with the number of wavelengths. The wavelength-dependent albedo $a(\lambda)$ and asymmetry $g(\lambda)$ come from the same opacity table (λ, a, $\langle\cos\rangle$, $C_{\rm ext}/{\rm H}$).

5.2 Wavelength sampling and packet luminosity

The source spectrum L_λ — a Planck function at `par%tstar`, a tabulated `par%source_spectrum`, or a spectrum for each component in the multi-source case — is integrated over each bin to a luminosity fraction $f_i = L_i/L_{\rm tot}$, and the birth bin is drawn from $\{f_i\}$ with an alias table. Each packet then carries a physical luminosity $L_{\rm p} = L_{\rm tot}/N_{\rm phot}$, so that the sum of packet luminosities reproduces the source luminosity and the downstream emission budget is absolute (erg s^{-1}). Multiple stellar populations (§10.) are mixed by drawing the emitting component in proportion to its luminosity.

The source spectrum may be supplied in two ways. As a *shape*, the file is renormalized and the absolute scale comes from `par%luminosity`. As an *absolute* (physical) spectrum, the source luminosity is instead the integral of the tabulated L_λ over the wavelength grid,

$$L_{\rm tot} = \int L_\lambda\,{\rm d}\lambda, \tag{12}$$

and inputs tabulated in frequency or energy are converted with the Jacobians

$$L_\lambda = L_\nu\,\frac{c}{\lambda^2}, \qquad L_\lambda = L_E\,\frac{hc}{\lambda^2}. \tag{13}$$

An isotropic external radiation field is specified instead by its mean intensity J; the luminosity it injects across an illuminated boundary of area $A_{\rm surface}$ is

$$L_{\rm tot} = \pi J A_{\rm surface}, \tag{14}$$

after which the packets are drawn and weighted exactly as for an internal source.

Composed sources. An internal source (or sources) and an external isotropic field may coexist in one run. Each packet is then assigned an origin — internal or external — with probability $L_{\rm int} : L_{\rm ext}$, and all packets share the luminosity

$$L_{\rm p} = \frac{L_{\rm tot}}{N_{\rm phot}}, \qquad L_{\rm tot} = L_{\rm int} + L_{\rm ext} = \sum_i L_i + \pi J A_{\rm surface}, \tag{15}$$

so the absolute normalization is the sum of the internal luminosity $L_{\rm int} = \sum_i L_i$ (Eq. 14 with $A_{\rm surface}$ the illuminated boundary) and the injected external luminosity. Because energy is additive, the composed image equals the internal-only image plus the external-only image, and the output is normalized to $L_{\rm tot}$.

5.3 Quasi-random photon launching

The launch of a packet is a fixed, low-dimensional integral over the unit hypercube: the origin (internal or external), the emitting component, the wavelength bin, the emission (or incidence) direction, and, for external illumination, the entry point on the boundary. With `par%launch_sequence ='sobol'` these coordinates are taken from an Owen-scrambled Sobol sequence instead of independent pseudo-random numbers. A low-discrepancy point set fills the launch cube more uniformly: for a smooth d-dimensional integrand the quadrature error can approach $N^{-1}(\log N)^d$ instead of the Monte-Carlo $N^{-1/2}$, and even for the discontinuous transport integrand the stratification is retained where it matters most. In particular, inverting one stratified coordinate through the monotonic luminosity CDF makes the packet count in a wavelength bin of probability p deviate from pN by order one rather than binomially (the alias table used by the pseudo-random path is replaced by an inverse CDF for this coordinate, because the alias permutation of the unit interval destroys the monotonicity that stratification requires).

The design is deliberately hybrid. Only the launch coordinates — a fixed seven-dimensional layout whose meanings never depend on the source configuration — come from the scrambled net, indexed by the global photon number carried as `photon%id`; every conditional, variable-length draw after launch (forced first scattering, the phase function, free paths) stays on the Mersenne Twister, so the transport estimator is unchanged and unbiased. Owen scrambling (nested-uniform hashing) preserves the net's equidistribution while making replicates with different `par%qmc_seed` values statistically independent, which restores an ordinary error estimate from the replicate-to-replicate scatter. Because the launch set is indexed globally, it is independent of the MPI task count; the direct images of internal sources, which depend on the launch variables alone, are then bit-identical for any number of ranks. The scattered field keeps its $N^{-1/2}$ behavior: the benefit concentrates in the wavelength allocation, the direct images, and the entry-point uniformity of external illumination.

6. The radiation field: Lucy path-length estimator

6.1 Mean intensity

Dust heating is set by the mean intensity (zeroth angular moment of I_λ) in each cell,

$$J_\lambda(\mathbf{r}) = \frac{1}{4\pi} \int I_\lambda(\mathbf{r}, \hat{\mathbf{n}})\, \mathrm{d}\Omega. \tag{16}$$

MoCafe estimates J_λ with the Lucy (1999) *path-length* estimator. The energy density is $u_\lambda = (4\pi/c)J_\lambda$, and the total path length that radiation traverses in a volume V per unit time equals c times the number of photons in V; accumulating a contribution from every segment a packet takes through the cell therefore estimates J_λ directly:

$$J_\lambda = \frac{1}{4\pi V\, \Delta\lambda\, d_{\rm cm}^2} \sum_{\text{segments in cell}} L_{\rm p}\, w\, \ell, \tag{17}$$

where ℓ is the segment length, V the cell volume, $\Delta\lambda$ the bin width, and $d_{\rm cm}$ the length unit in cm.

6.2 Path-length versus collision estimator

A collision (or absorption) estimator would tally only at discrete interaction points, so a cell with few interactions receives a noisy, or zero, estimate. The path-length estimator instead credits every fly-through, whether or not the packet interacts, so even optically thin cells — where interactions are rare but many packets pass through — get a smooth, low-variance J_λ. This is decisive for dust heating, where the illuminating field must be known in cells that are optically thin to the starlight.

6.3 Absorbed power and the analytic first flight

The power absorbed in a cell follows from J_λ and the absorption opacity,

$$P_{\rm abs} = 4\pi V \int \kappa_{\rm abs}(\lambda) J_\lambda\, \mathrm{d}\lambda. \tag{18}$$

The forced first flight (§5.) has a biased free-path distribution, so that segment is *not* tallied by sampling; instead its exact expectation is added analytically along the ray to the edge,

$$J_\lambda \mathrel{+}= \frac{w L_{\rm p}}{4\pi V \Delta\lambda\, d_{\rm cm}^2} \int e^{-\tau_\lambda(\ell)}\, \mathrm{d}\ell = \frac{w L_{\rm p}}{\cdots}\, \frac{e^{-s\,\tau_{\rm in}} - e^{-s\,\tau_{\rm out}}}{\kappa_{\rm ext} s}, \tag{19}$$

a zero-variance direct component. Flights with $n_{\rm scatt} \geq 1$ sample the ordinary exponential free path and are tallied along the walked segments. The `_jlam` output reports the pathlength-tallied absorbed power (`EABS_A`) against an independent event counter (`EABS_B`) as an energy-conservation check.

7. Dust thermal emission I: Lucy + SEDust

The radiation field J_λ from Eq. (17) is handed cell by cell to the SEDUST grain library, which returns the emission spectrum for the local field. Below, $C_{\rm abs}(a, \lambda) = \pi a^2 Q_{\rm abs}(a, \lambda)$ is the absorption cross section of a grain of radius a.

7.1 Grain energy balance

A grain absorbs starlight at the rate $\dot{E}_{\rm abs} = \int 4\pi C_{\rm abs}(a,\lambda) J_\lambda \, d\lambda$ and radiates at $\dot{E}_{\rm emit}(T) = \int 4\pi C_{\rm abs}(a,\lambda) B_\lambda(T) \, d\lambda$. A *large* grain reaches a steady temperature $T_{\rm eq}$ where $\dot{E}_{\rm emit}(T_{\rm eq}) = \dot{E}_{\rm abs}$; because the absorbed energy per photon is tiny compared with the grain heat capacity, its temperature barely fluctuates. A *small* grain (or a PAH molecule), by contrast, has so small a heat capacity that a single ultraviolet photon can raise its temperature by hundreds of kelvin, after which it cools before the next hit. Its temperature is therefore a random variable, and the emission must be integrated over its probability distribution.

7.2 Enthalpy and heat capacity

The internal energy (enthalpy) $U(T)$ of a grain is obtained from the Debye model of the lattice vibrations (Draine & Li 2001): silicate and graphite use two- and three-dimensional Debye integrals with the appropriate Debye temperatures, and PAHs add discrete C–H and C–C mode contributions (`enthalpy_v2.f90`, `enthalpy_astrodust.f90`). Inverting $U(T)$ gives the temperature reached after absorbing a given energy, which is the input to the stochastic-heating solver.

7.3 Stochastic heating: the temperature probability distribution

SEDUST solves for the steady-state probability distribution P_j over a set of discrete enthalpy bins $\{U_j\}$ ($j = 1 \ldots N$, default $N = 200$) using Draine's thermal-discrete method (`stoch_qm.f90`; Guhathakurta & Draine 1989; Draine & Li 2001). Transitions between bins are collected into a rate matrix A_{fi} (the rate of going from bin i to bin f):

- **Heating** ($f > i$): absorption of a photon of energy $U_f - U_i = hc/\lambda$ carries the grain upward, at rate

$$A_{fi} = \frac{4\pi J_\lambda}{hc/\lambda} C_{\rm abs}(a,\lambda) \frac{d\lambda}{dU_f}, \tag{20}$$

  i.e. proportional to the photon arrival rate at the wavelength that bridges the two bins.

- **Cooling** ($f < i$): thermal emission removes energy. In the thermal-discrete treatment (`dbdis`, the production setting) all downward transitions are retained, with rates set by the Planck emission at the temperature of the upper bin; the continuous-cooling approximation (`dbcon`) collapses these into nearest-neighbour ($f = i-1$) transitions, recovering the Guhathakurta–Draine temperature-space recursion.

Conservation of probability fixes the diagonal, and the steady state satisfies

$$\sum_{i \neq j} A_{ji} P_i = P_j \sum_{i \neq j} A_{ij}, \qquad \sum_j P_j = 1, \tag{21}$$

solved as a sparse linear system with a bi-conjugate-gradient solver. The result P_j is the probability that the grain sits in enthalpy bin j, i.e. at temperature $T_j = T(U_j)$.

7.4 Equilibrium limit

For large grains the distribution P_j collapses to a narrow spike at $T_{\rm eq}$, and the solve reduces to the energy-balance root $\dot{E}_{\rm emit}(T_{\rm eq}) = \dot{E}_{\rm abs}$. SEDUST detects this regime and can use the equilibrium temperature directly, which is exact for the big grains that dominate the far-infrared and much cheaper than the full solve.

7.5 Size distribution and PAHs

The cell emissivity integrates the single-grain emission over the grain-size distribution $\mathrm{d}n/\mathrm{d}a$ of each material (silicate, carbonaceous, PAH),

$$\varepsilon_\lambda = \sum_{\text{species}} \int \mathrm{d}a \, \frac{\mathrm{d}n}{\mathrm{d}a} \, 4\pi C_{\text{abs}}(a,\lambda) \int \mathrm{d}T \; P(a,T) B_\lambda(T), \tag{22}$$

where $P(a,T)$ is the stochastic distribution above (a delta function at T_{eq} for large grains). The PAH population produces the mid-infrared band spectrum (3.3, 6.2, 7.7, 8.6, 11.3 μm); the smallest grains produce the smoothly rising continuum from single-photon heating. The size distributions and optical constants come from the selected grain model.

7.6 Emission spectrum and energy normalization

The emission in each cell is renormalized so that its wavelength integral equals the locally absorbed power of Eq. (18),

$$\int \varepsilon_\lambda^{\text{norm}} \, \mathrm{d}\lambda = P_{\text{abs}}, \tag{23}$$

which makes energy conservation *exact* regardless of the internal SEDUST normalization. The normalized spectrum is resampled onto the transport wavelength grid and either integrated directly (the intrinsic dust SED) or emitted as a second batch of packets — sampled in wavelength from $\varepsilon_\lambda^{\text{norm}}$ and in position from the cell emissivity — to form the emergent SED and the pixel-resolved emission image.

7.7 Speed options

The exact stochastic solve costs $\mathcal{O}(0.1\,\text{s})$ per cell. Two approximations trade accuracy for speed:

`par%dust_single_teq` replaces $P(a,T)$ by a single mixture equilibrium temperature. It keeps the total power but drops the PAH and stochastic features (no mid-infrared bands), and is intended for optically thin, far-infrared-dominated problems.

`par%dust_fast_table` exploits that, to good approximation, the emission spectrum depends on the field only through its intensity $U = \int J_\lambda \, \mathrm{d}\lambda / \int J_\lambda^{\text{ref}} \, \mathrm{d}\lambda$ and shape. A reference emission table is built over a grid of U once, and each cell is interpolated in $\log U$. This preserves the PAH/stochastic features at $\sim 1\%$ SED-shape error, with the largest deviations on the far-infrared slope.

7.8 Self-absorption iteration

At high infrared optical depth the dust re-absorbs its own emission. With `par%dust_niter`> 1 MoCafe transports the emitted packets, tallies their contribution to the field, adds it to the stellar field $J_\lambda \to J_\lambda^\star + J_\lambda^{\text{dust}}$, and recomputes the emission, iterating until the total emitted luminosity converges to `par%dust_tol`. Writing f for the fraction of dust emission that is re-absorbed, the emitted luminosity obeys

$$L^{(k)} = L_{\text{abs}}^\star + f\, L^{(k-1)} \;\longrightarrow\; \frac{L_{\text{abs}}^\star}{1-f}, \tag{24}$$

a geometric series. At $\tau_V = 100$ this converges in about ten iterations, and only the converged run conserves energy: a single pass loses the re-absorbed dust emission (only $\sim 37\%$ of the starlight escapes), whereas the converged run returns 99.8%. Iteration is therefore *required* for energy conservation at high infrared optical depth.

8. Dust thermal emission II: Bjorkman & Wood

The Bjorkman & Wood (2001) scheme uses fixed-energy packets and a mixture mean opacity, and needs no radiation-field storage and no iteration. When a packet is absorbed in a cell, the cell's cumulative absorbed energy increases, raising its equilibrium temperature from T to $T+\Delta T$; the packet is re-emitted *immediately* with a wavelength drawn from the temperature-correction spectrum. The construction is as follows. A cell that has already emitted a spectrum $\propto \kappa_{\rm abs} B_\lambda(T)$ must, after the temperature increment, radiate $\propto \kappa_{\rm abs} B_\lambda(T+\Delta T)$. The *additional* photons needed to correct the spectrum are therefore distributed as the difference,

$$\frac{{\rm d}P}{{\rm d}\lambda} \;\propto\; \kappa_{\rm abs}(\lambda)\left[B_\lambda(T{+}\Delta T) - B_\lambda(T)\right] \;\approx\; \kappa_{\rm abs}(\lambda)\,\frac{\partial B_\lambda(T)}{\partial T}, \tag{25}$$

which is the sampling law for the reemitted packet. Summed over all absorptions this makes each cell radiate the correct $\kappa_{\rm abs} B_\lambda(T)$ spectrum in steady state. The equilibrium temperature is found by inverting the monotonic $\kappa_{\rm P}(T) = \int \kappa_{\rm abs}(\lambda)\,B_\lambda(T)\,{\rm d}\lambda$ against the running absorbed power. The method is approximate (equilibrium only, mixture opacity, no PAH/stochastic features) but cheap, and the emergent dust light lands directly in the observer `Scattered` SED. Over $\tau_V = 0.1\ldots20$ its absorbed luminosity agrees with the Lucy method to $< 2\%$.

9. Modified Random Walk

In cells that are very optically thick to *scattering*, a packet would take an enormous number of tiny steps. The Modified Random Walk (Min et al. 2009; Robitaille 2010) replaces a whole diffusive excursion with one step. A sphere of radius R_0 (the distance to the nearest cell wall) is inscribed at the packet position; MRW triggers when its scattering optical thickness exceeds a threshold, $\kappa_{\rm sca} R_0 > \gamma$ (`par%mrw_gamma`, default 2). The dimensionless first-passage variable y — the probability that the diffusion has not yet reached the sphere surface — has cumulative distribution

$$P(y) = 2\sum_{n=1}^{\infty}(-1)^{n+1}\,y^{n^2}, \qquad y \in [0,1], \tag{26}$$

which MoCafe tabulates and inverts once at setup. A draw of y fixes the total *scattering* path length travelled inside the sphere,

$$ct = -\ln y \;\cdot\; 3\,\kappa_{\rm sca}\left(\frac{R_0}{\pi}\right)^2, \tag{27}$$

after which the packet is moved to a random point on the sphere with a fresh isotropic direction. MRW diffuses with the *scattering* opacity; the absorption along the path is applied as a weight decay $w \leftarrow w\,e^{-\kappa_{\rm abs}\,ct}$, and the corresponding absorbed energy is deposited into the radiation-field tally as the path integral $\int_0^{ct} w\,L_{\rm p}\,e^{-\kappa_{\rm abs}\ell}\,{\rm d}\ell$. Russian roulette terminates high-albedo, high-τ excursions. Because realistic dust is genuinely absorbing, the diffusion is usually terminated by absorption before MRW helps; its speed-up is large only for near-pure-scattering, very thick cells, so MRW is off by default and adds no measurable overhead when it rarely triggers.

10. Source (galaxy) models

A stellar component emits packets from a spatial distribution; multiple components (`par%nsource`) are selected in proportion to luminosity, and each may have its own spectrum and geometry (bulge + disk gives the bulge-to-total ratio).

10.1 Exponential disk

A razor-thin exponential disk has surface density $\Sigma(r) \propto e^{-r/r_s}$, so the probability of emitting from cylindrical radius r is

$$p(r)\,\mathrm{d}r \propto 2\pi r\,\Sigma(r)\,\mathrm{d}r \propto r\,e^{-r/r_s}\,\mathrm{d}r, \tag{28}$$

the Gamma(2) (double-exponential) law. It is sampled *exactly* as the sum of two exponential deviates, $r = -r_s(\ln\xi_1 + \ln\xi_2)$ (`rand_r1exp`), avoiding any inverse-CDF search. The azimuth is uniform (modified by spiral arms, below).

10.2 Vertical structure

The vertical profile is exponential, $\rho \propto e^{-|z|/z_s}$ (`rand_zexp`), or the self-gravitating isothermal-sheet form $\rho \propto \mathrm{sech}^2(z/z_s)$ (`rand_sech2`), selected by `par%density_zprofile`. The same options apply to the dust disk, whose density is the double exponential $\exp(-|z|/z_s)\exp(-r/r_s)$.

10.3 Spiral arms

Log-spiral arms multiply the disk by

$$1 + A\,\sin\left[m\left(\frac{\ln r}{\tan p} - \phi\right)\right], \tag{29}$$

with m arms, pitch angle p, and amplitude A (`par%spiral_m`, `par%spiral_pitch`, `par%spiral_amp`). For *sources* the azimuth ϕ is drawn by rejection against this factor (accept if $(1+A)\xi \le 1 + A\sin[\ldots]$); for the *dust* density the cell opacity is multiplied by the same factor. Seen face-on, the absorbed-power (or dust-emission) map traces the arms, validated in `examples/galaxy/spiral_faceon.in`.

10.4 Sersic bulge: deprojection and alias sampling

The projected Sersic surface-brightness profile is $I(R) = I_0\exp[-b_m(R/R_e)^{1/m}]$ with index m (`par%src_sersic_index`), effective radius R_e (`par%src_reff`), and b_m from Ciotti & Bertin (1999) so that R_e encloses half the projected light. MoCafe emits from the *three-dimensional* luminosity density $\nu(r)$, recovered from $I(R)$ by the inverse Abel transform,

$$\nu(r) = -\frac{1}{\pi}\int_r^\infty \frac{\mathrm{d}I}{\mathrm{d}R}\,\frac{\mathrm{d}R}{\sqrt{R^2 - r^2}}. \tag{30}$$

The deprojected cumulative luminosity within spherical radius r is

$$\mathcal{L}(<r) = \gamma\big(2m{+}1,\, b_m r^{1/m}\big) + \frac{2}{\pi}\,\frac{b_m^{2m+1}}{m\,\Gamma(2m{+}1)}\,r^2\int_1^\infty x\,f(x)\,e^{-b_m(rx)^{1/m}}(rx)^{1/m}\,\mathrm{d}x, \tag{31}$$

computed once on a logarithmic radius grid (`sersic_cumulative_3D`, with the incomplete gamma function γ). The radius is then drawn with the *alias* method (§11.) rather than an inverse-CDF search: the bin probabilities (differences of the cumulative) build an alias table, and `rand_alias_linear` returns a continuous radius by linear interpolation within the selected bin. The point is placed isotropically and flattened in z by the axial ratio (`par%src_axial_ratio`). The alias-sampled distribution reproduces the analytic cumulative to $\sim 10^{-4}$.

10.5 Boxy / bar / x-bar bulges

Generalized-ellipsoid bulges use the generalized radius $r = (|x|^n + |y|^n + |z|^n)^{1/n}$ with boxiness n (`par%src_boxiness`; $n = 2$ is an ellipsoid, $n > 2$ boxy) and a power-law (`boxy`), Gaussian (`bar`), or peanut (`xbar`) radial profile. These are drawn with a slice-within-Gibbs sampler (`random_bulge.f90`): an auxiliary height is drawn uniformly under the current density value, and each coordinate is then updated uniformly on the resulting horizontal slice. Consecutive draws are correlated (the sampler explores by conditional updates), which is adequate for a smooth bulge but statistically weaker than the independent alias-sampled Sersic.

11. Random sampling

11.1 Inversion and rejection

Two elementary methods recur above. *Inversion* solves $P(x) = \xi$ for the cumulative P; it is exact and consumes one deviate, used for the free path (Eq. 4), the Henyey–Greenstein cosine, and the exponential disk. *Rejection* proposes x from an envelope and accepts with probability proportional to the target density; it needs no cumulative and is used for the spiral azimuth and inside the Gibbs bulge sampler, at the cost of occasional rejected proposals.

11.2 Alias method

Walker's alias method samples an index from a discrete distribution $\{p_k\}$ ($k = 1 \ldots K$) in $\mathcal{O}(1)$ time after an $\mathcal{O}(K)$ setup. The setup (`random_alias_setup`, Vose's construction) rewrites the distribution as a set of K equal-probability "columns," each holding at most two outcomes: it stores an acceptance probability q_k and an alias A_k for every bin. A draw picks a uniform bin k and a uniform $\eta \in [0, 1)$, and returns k if $\eta < q_k$, else A_k. MoCafe uses it for wavelength sampling (§5.), the Mueller S_{11} angle, and the Sersic radius — anywhere a fixed discrete distribution is sampled many times.

11.3 Continuous piecewise-linear alias

For a continuous variable tabulated at nodes x_i with density p_i, the probability of the interval $[x_i, x_{i+1}]$ is the trapezoid area $\frac{1}{2}(p_i + p_{i+1})(x_{i+1} - x_i)$. An alias table over these interval probabilities selects a bin in $\mathcal{O}(1)$; within the bin the value is returned by inverting the linear density (`rand_alias_linear`), giving a continuous, independent draw. This is used for the Sersic radius and for external-illumination angles.

12. HEALPix all-sky interior observer

For an observer *inside* the medium (the Milky-Way case) there is no single image plane. Each scattering, dust-reemission, and direct event peels toward the observer point, and the expected contribution (Eq. 8, with the $1/d_o^2$ replaced by the pixel solid angle) is binned into the HEALPix pixel of the arrival direction $\hat{\mathbf{n}}$. HEALPix (Górski et al. 2005) tiles the sphere into $n_{\rm pix} = 12\,n_{\rm side}^2$ equal-area pixels of solid angle $\Omega = 4\pi/n_{\rm pix}$; the RING scheme is used, and the pixel index is found by `vec2pix`. The optical depth to the observer is integrated along the matching ray (Cartesian or AMR). The output carries the standard HEALPix keywords so it reads directly with `healpy`. Because each event maps to the single pixel of its *source* direction, a diffuse map needs many events distributed over the sky; concentrated (inner-disk) far-infrared emission is sparse at high $n_{\rm side}$ and benefits from a finer spatial grid or smoothing.

13. Energy conservation and validation

Energy conservation is the main internal check and holds by construction:

- The total stellar luminosity equals the escaped starlight plus the absorbed (hence re-emitted) power.
- The dust solvers normalize the emission to the absorbed power cell by cell, so the integrated intrinsic dust luminosity equals `L_ABS` to machine precision, and the emergent SED integrated over $4\pi d^2$ recovers the same value to Monte-Carlo accuracy.
- The `_jlam` output cross-checks the path-length-tallied absorbed power against an independent event counter.

Against external references, the SEDUST single-cell emission tracks the Camps et al. (2015) seven-code stochastic-heating benchmark (which includes SKIRT and DIRTY) to a 2–4% median relative difference over $U = 10^{-2}\ldots10^{4}$, the Lucy and Bjorkman & Wood absorbed luminosities agree to $< 2\%$ over $\tau_V = 0.1\ldots20$, and dust emission on a uniform octree sphere reproduces the equivalent Cartesian sphere. These are consolidated in the `examples/benchmarks/validate.py` regression harness.

14. Summary of key numerical parameters

Table 1: Parameters that control the algorithms in this note.

Parameter	Meaning	Default
`par%use_reduced_wgt`	forced first scattering + reduced weight	`.true.`
`par%use_stokes`	polarized Mueller-matrix scattering	`.false.`
`par%lambda_ref`	reference wavelength for grey rescaling	—
`par%dust_model_sed`	grain model (`astrodust`/`dl07`/`zubko`)	`astrodust`
`par%dust_single_teq`	equilibrium-only emission (fast)	`.false.`
`par%dust_fast_table`	U-interpolated emission table	`.false.`
`par%dust_niter`	self-absorption iterations	1
`par%dust_tol`	iteration convergence tolerance	—
`par%mrw_gamma`	MRW trigger ($\kappa_{\rm sca}R_0 > \gamma$)	2
`par%nside`	HEALPix resolution (interior observer)	—

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